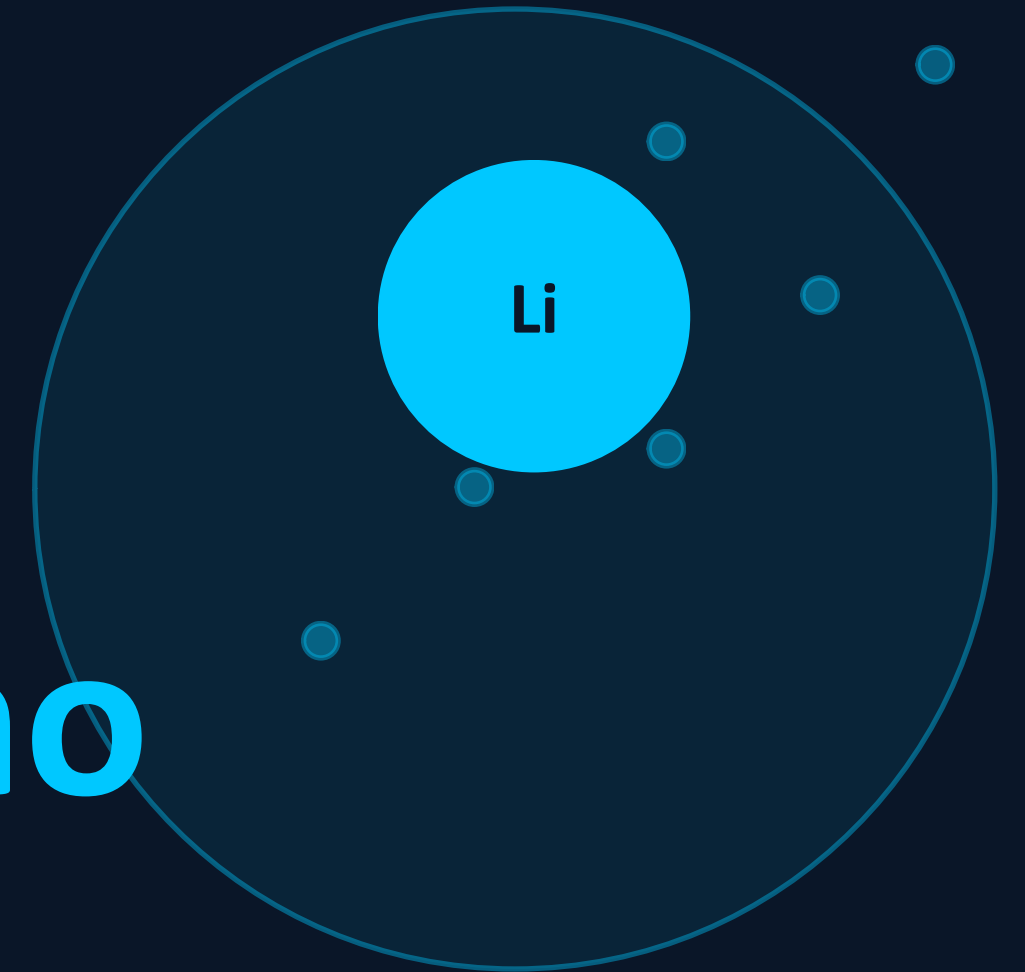


ADC MINI PROJECT

Li-Fi Based Data Transmission using Arduino



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What is Li-Fi?

- Li-Fi stands for Light Fidelity
- Uses visible light for wireless data transmission
- Coined by Prof. Harald Haas in 2011
- Uses LED bulbs as access points
- Data encoded by rapid light intensity changes
- Operates over the visible light spectrum (400–700 THz)

Li-Fi vs Wi-Fi		
Medium	Light	Radio
Speed	Up to 224 Gbps	Up to ~10 Gbps
Security	High	Moderate
Range	Short (LOS)	Long (NLOS)
Interference	None (RF)	Yes (RF)

Objective of the Project

03

Goals & Scope

01

Demonstrate Li-Fi Communication

Implement a working Li-Fi system that transmits data using visible light from a smartphone flashlight.

02

Arduino as Decoder

Use Arduino UNO to receive light signals via an LDR sensor, decode binary data, and process messages.

03

LCD Display Output

Display decoded text messages on a 16×2 LCD display in real time, proving successful data reception.

04

Explore Practical Feasibility

Evaluate Li-Fi as a cost-effective, RF-interference-free alternative to traditional wireless communication.

Components Used

01 Arduino NANO

Microcontroller board — core processing unit

02 LDR Sensor

Light Dependent Resistor — detects light signals

03 16×2 LCD Display

Alphanumeric display for decoded message output

04 Resistors

220Ω & 10kΩ for current limiting and pull-down

05 Breadboard

Prototyping board for circuit connections

06 Jumper Wires

Male-to-male wires for breadboard connections

07 Smartphone + App

Li-Fi transmitter app controls flashlight signals

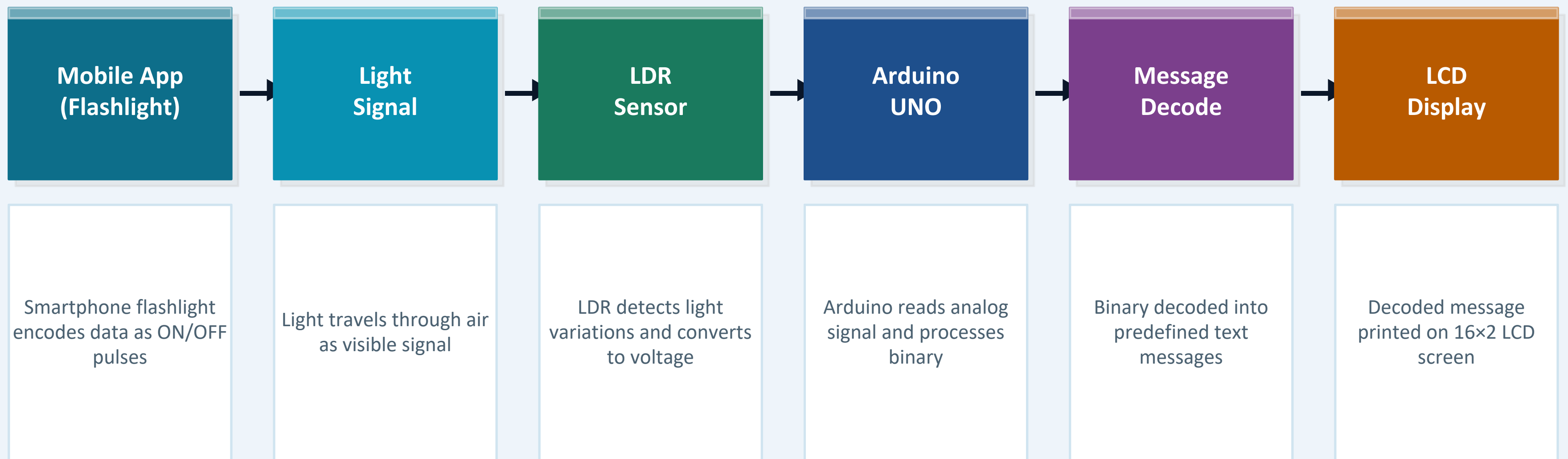
08 USB Cable

Powers and programs the Arduino UNO board

Block Diagram

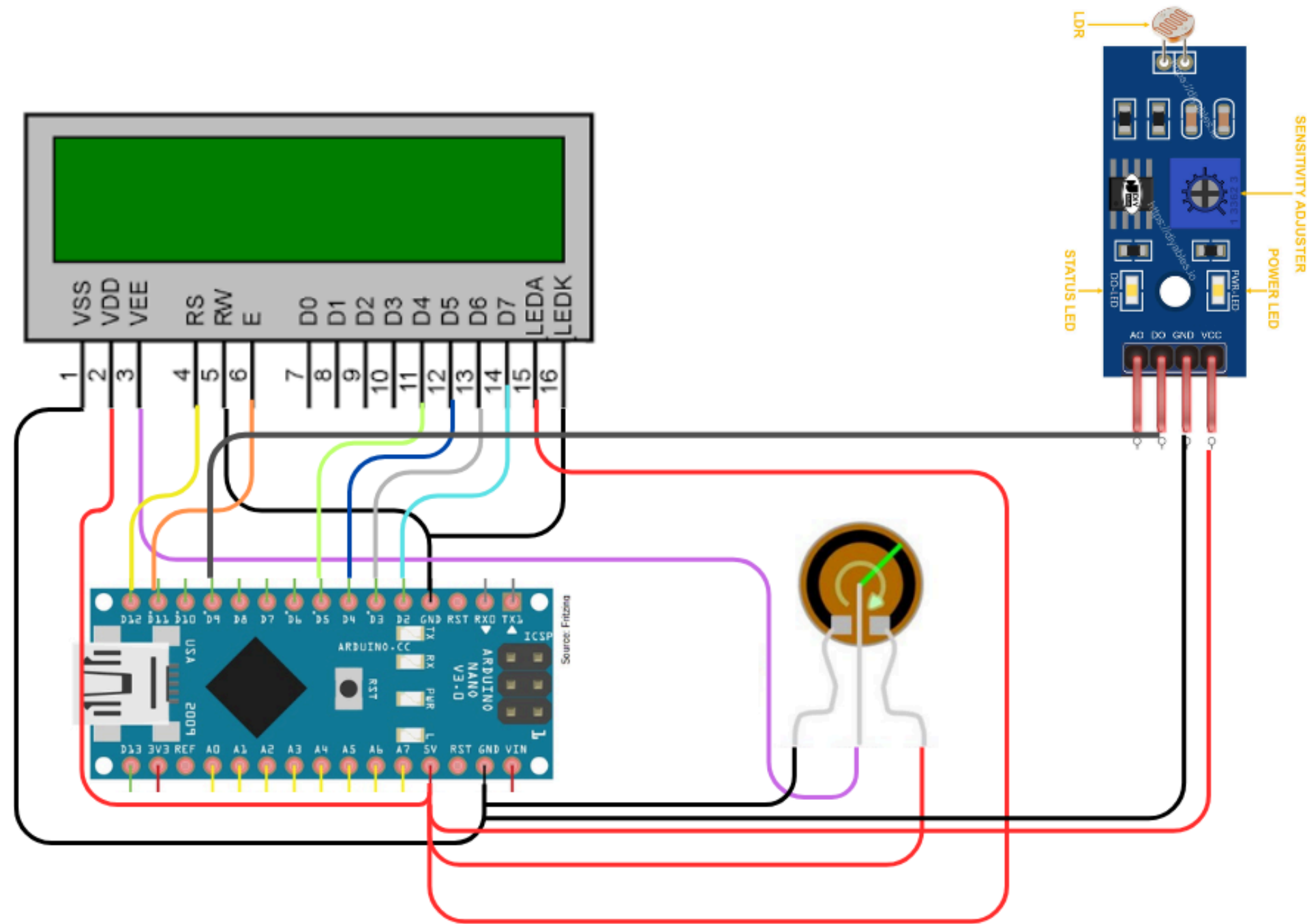
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System Data Flow



Circuit Diagram

04



1

Signal Generation

The smartphone flashlight app rapidly turns the LED ON and OFF to encode data as binary light pulses.

2

Signal Detection

The LDR (Light Dependent Resistor) detects changes in light intensity and generates corresponding voltage variations.

3

Analog to Digital

Arduino reads analog values from the LDR via the A0 pin and compares against a threshold to identify 0s and 1s.

4

Decoding & Display

Arduino decodes the binary sequence into predefined messages and outputs them on the 16x2 LCD display.

Key Facts

Sampling Rate ~1000 Hz

LDR Threshold Analog > 512

Binary Bits 8-bit encoding

Baud Rate Low (app-defined)

Range ~1–2 metres

Power 5V from USB

● ● ● Arduino Code Snippet

```
#include <LiquidCrystal.h>

// LCD pins: RS, E, D4, D5, D6, D7
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

#define LDR 9

String data = "";
String lastMessage = "";

void setup() {
  Serial.begin(9600);
  pinMode(LDR, INPUT);

  lcd.begin(16,2);

  lcd.clear();
  lcd.print("LiFi Project");
  delay(2000);

  lcd.clear();
  lcd.print("Send message");
  lcd.setCursor(0,1);
  lcd.print("from LiFi App");
  delay(2000);

  lcd.clear();
  lcd.print("Waiting...");
}

void loop() {
  int val = digitalRead(LDR);

  if(val == LOW) { // light detected
    data += "0";
  }
  else { // no light
    data += "1";
  }

  Serial.println(data);

  String msg = decodeMessage(data);

  if(msg != "" && msg != lastMessage) {
    lastMessage = msg;

    lcd.clear();
    lcd.print(msg);

    Serial.print("Received: ");
    Serial.println(msg);

    data = "";
  }

  if(data.length() > 15) {
    data = "";
  }

  delay(100);
}

String decodeMessage(String d) {
  if(d == "001") return "hi";
  if(d == "0001") return "hello";
  if(d == "00001") return "how are you";
  if(d == "000001") return "I am fine";
  if(d == "0000001") return "ok";
  if(d == "00000001") return "good morning";
  if(d == "000000001") return "good afternoon";
  if(d == "0000000001") return "good evening";
  if(d == "00000000001") return "thank you";
  if(d == "000000000001") return "sorry";

  return "";
}
```

①

Read LDR Input

analogRead() samples LDR voltage on pin A0 every millisecond

②

Threshold Check

Values above 512 = light ON (binary 1); below = light OFF (binary 0)

③

Bit Accumulation

8 bits collected in sequence form one ASCII character byte

④

Decode & Display

Byte matched to predefined messages; output sent to LCD via I2C

Applications of Li-Fi

08

Smart Homes

LED lighting doubles as a data network — internet access through every light bulb without extra wiring.

Hospitals

RF-free data transfer in ICUs and ORs where Wi-Fi signals can interfere with medical equipment.

Underwater Comms

Light travels in water better than radio waves — enables diver communication and submarine systems.

Aircraft Cabins

In-flight internet via LED cabin lights, eliminating need for costly RF infrastructure on planes.

Military

Highly secure, undetectable line-of-sight communication ideal for covert military operations.

Smart Cities

Street LED lamps provide both lighting and broadband data access for IoT-connected infrastructure.

Advantages & Limitations

09

✓ Advantages

- Extremely high data speeds (up to 224 Gbps)
- No radio frequency interference
- Highly secure — light does not pass through walls
- Energy efficient — uses existing LED infrastructure
- Ideal for RF-sensitive environments (hospitals, aircraft)
- Virtually unlimited bandwidth in visible light spectrum

⚠ Limitations

- Requires direct line-of-sight (LOS) between devices
- Cannot penetrate walls — limited to one room
- Affected by strong ambient light interference
- Short operational range (~1–10 meters typically)
- Receiver must always be in the light path
- Still in early-stage commercial deployment globally

CONCLUSION

Li-Fi: Illuminating the Future of Wireless Communication

- Successfully demonstrated Li-Fi data transmission using Arduino and LDR sensor
- Proved that visible light can reliably carry encoded digital messages
- Li-Fi offers a secure, high-speed, RF-free alternative to conventional Wi-Fi
- With further development, Li-Fi has immense potential in smart cities, healthcare, and beyond

224

Gbps · Max Speed

5V

USB · Power Source

400–700

THz · Light Spectrum